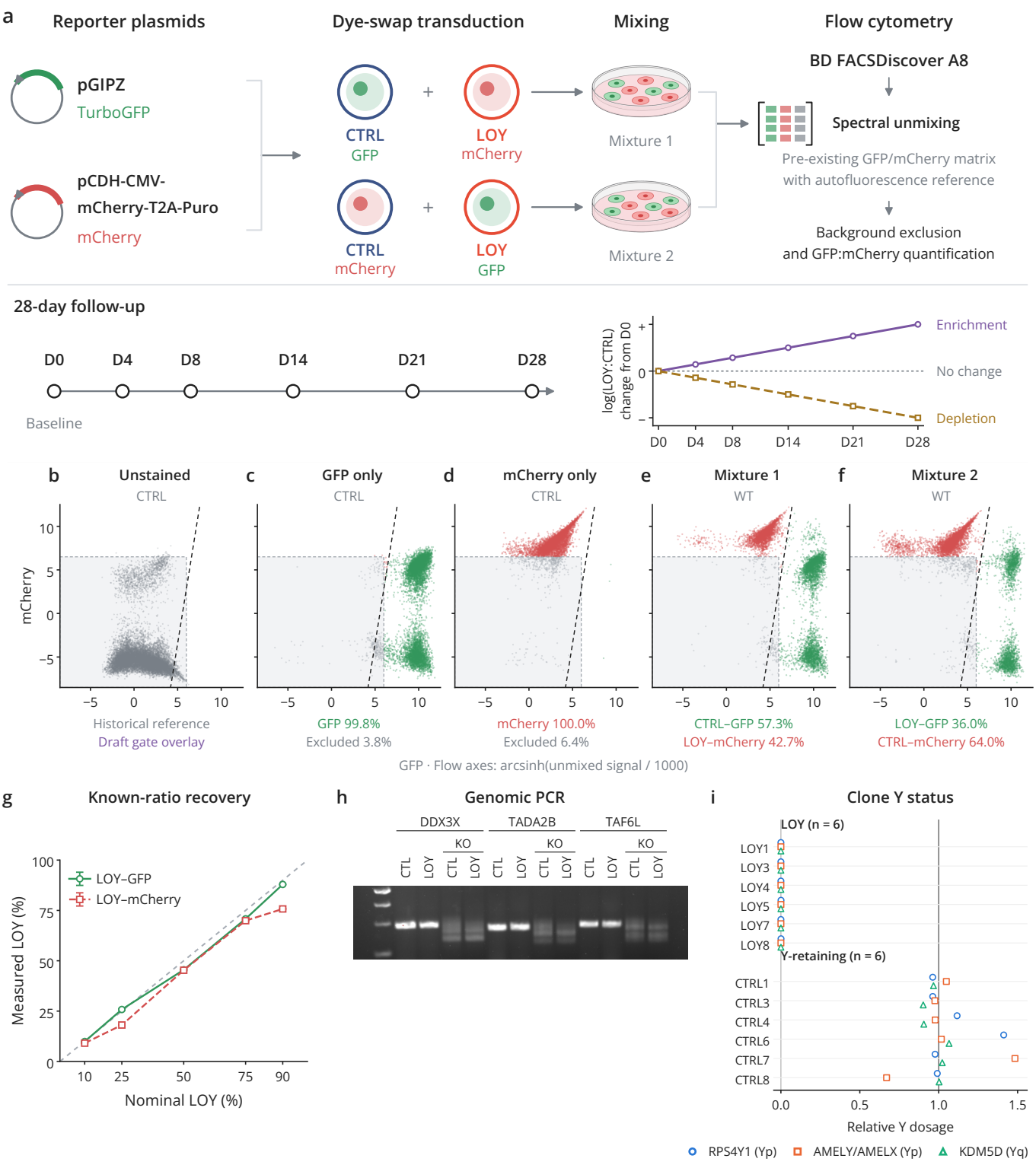

a
Reporter plasmids
Dye-swap transduction
Mixing
Flow cytometry
pGIPZ
TurboGFP
pCDH-CMV-
mCherry-T2A-Puro
mCherry
CTRL
GFP
LOY
mCherry
CTRL
mCherry
LOY
GFP
Mixture 1
Mixture 2
BD FACSDiscover A8
Spectral unmixing
Pre-existing GFP/mCherry matrix
with autofluorescence reference
Background exclusion
and GFP:mCherry quantification
28-day follow-up
D0
D4
D8
D14
D21
D28
Baseline
log(LOY:CTRL)
change from D0
Enrichment
No change
Depletion
b
Unstained
CTRL
c
GFP only
CTRL
d
mCherry only
CTRL
e
Mixture 1
WT
f
Mixture 2
WT
mCherry
Historical reference
Draft gate overlay
GFP 99.8%
Excluded 3.8%
mCherry 100.0%
Excluded 6.4%
CTRL–GFP 57.3%
LOY–mCherry 42.7%
LOY–GFP 36.0%
CTRL–mCherry 64.0%
GFP · Flow axes: arcsinh(unmixed signal / 1000)
g
Known-ratio recovery
LOY–GFP
LOY–mCherry
Measured LOY (%)
Nominal LOY (%)
h
Genomic PCR
DDX3X
TADA2B
TAF6L
KO
CTL
LOY
i
Clone Y status
LOY (n = 6)
LOY1
LOY3
LOY4
LOY5
LOY7
LOY8
Y-retaining (n = 6)
CTRL1
CTRL3
CTRL4
CTRL6
CTRL7
CTRL8
Relative Y dosage
RPS4Y1 (Yp)
AMELY/AMELX (Yp)
KDM5D (Yq)